Omar Khattab

Ottawa, ON omar.hosam2000@gmail.com +1 (613) 600-6921 github linkedin omarkhattab.ca

Summary

Computer Science graduate and OC Transpo Customer Information Systems specialist with bus and rail operations experience. Develops, integrates and maintains transit and business applications using web APIs and SQL databases. Combines GTFS data knowledge, Windows automation, software testing and user documentation with firsthand understanding of operational needs.

Education

University of Ottawa Sep 2020 – Apr 2026
Honours BSc in Computer Science; GPA: 7.5 / 10.0 *Ottawa, ON*

- **Relevant coursework:** Database Design, Software Engineering, Algorithms, Operating Systems, Network Security and Human-Computer Interaction.
- **Academic and personal projects:** Dental clinic system (Java/Spring/SQL), Android car-rental app, Python Discord bots, Java Huffman compression tool, Excel schedule generator (Python/openpyxl), Java client/server extensions and SciFact team data preparation and caching.

Professional Experience

OC Transpo – City of Ottawa Jun 2026 – Present
Business Specialist, Customer Information Systems (GTFS) *Ottawa, ON*

- Validate GTFS Static and GTFS-Realtime data for feed maintenance and service changes using various tools, including GTFS Validator and GTFS Viewer; built both applications to support feed inspection and issue investigation.
- Investigate discrepancies across Giro HASTUS exports, Clever Devices bus data and GTFS middleware, accounting for separate bus and rail paths and their effects on trip planners, information displays and developer feeds.
- Coordinate validation findings and service-change handoffs with planning staff and vendors; maintain activation checklists, troubleshooting procedures and reference documentation in SharePoint.

Independent Application Developer Jan 2020 – Present
Personal Projects and Client Applications *Ottawa, ON*

- Develop personal software tools and, more recently, transit and client applications; gather requirements and design, customize and maintain web interfaces, APIs and database components.
- Integrate APIs and transit data feeds with relational databases; manage schemas, migrations, triggers and access policies, and troubleshoot data and application issues.
- Manage code, configuration and automated test workflows in Git and GitHub; write maintenance procedures, clinic staff guides and Windows setup and recovery instructions.

OC Transpo – City of Ottawa Apr 2025 – Jun 2026
Electric Rail Operator *Ottawa, ON*

- Operated and monitored CBTC-equipped LRT under Control Centre direction; responded to alarms and service disruptions, documented incidents and escalations, and followed safety and operating procedures under time pressure.

OC Transpo – City of Ottawa Mar 2024 – Apr 2025
Bus Operator *Ottawa, ON*

- Operated OC Transpo routes and coordinated with the Control Centre during detours and emergencies; documented incidents and developed practical knowledge of schedules, stop assignments, reliefs and passenger information needs.

Technical Skills

Languages & Frameworks: Python, Java, JavaScript/TypeScript, C/C++, C#, VB.NET, SQL, PowerShell, Bash, HTML/XML; ASP.NET/.NET Framework, Node.js, Express, React, Next.js and Spring Boot/JDBC

Databases: SQL Server, Oracle SQL and PL/SQL (procedures and packages), PostgreSQL, MySQL, Microsoft Access, Supabase, Prisma, SQLite/Cloudflare D1 and Firebase; schemas, migrations and data integrity

Systems Design & Security: System Development Life Cycle (SDLC), structured systems analysis and design, object-oriented and event-driven design, n-tier architecture, REST APIs and Protocol Buffers; application security best practices, access controls and Win32/.NET interoperability

Testing & Configuration Management: Unit, integration, regression and load testing; test plans and test cases, Vitest, Python unittest, Node.js tests and Firebase emulators; Git, GitHub Actions, Docker, Vercel and Cloudflare

Transit & Business Systems: GTFS Static and GTFS-Realtime, HASTUS/CleverCAD exports, feed validation; Microsoft Office and SharePoint development; requirements analysis, troubleshooting, technical documentation and user guides

Data Analysis & Reporting: pandas, NumPy, Jupyter, statistics and Excel reporting

Selected Projects

GTFS Validator | *Python, SQLite, pandas, JavaScript, Java validator* GitHub

- Built a local Windows application that validates GTFS feed structure, schedule data and record relationships, and compares Giro HASTUS bus/rail and CleverCAD exports with final feeds to identify stop, trip, calendar and shape discrepancies.
- Integrated MobilityData's Java validator, disk-backed SQLite indexes and fingerprinted caches; added regression tests and Excel/printable findings to support repeatable validation and issue investigation.

Transit Detour Tracker and Reporting Dashboard OC Transpo – CIS

JavaScript, XML/RSS, Cloudflare Workers/D1, SQL, Node.js tests

- Built a CIS reporting dashboard combining OC Transpo XML/RSS alerts and webpage data with scheduled collection, historical SQL records and daily/monthly summaries, preserving publication and service-activation dates.
- Added authenticated collection, failure diagnostics, stale-data warnings and CSV exports; tested malformed XML, date boundaries and access denial, and preserved the last successful report during upstream failures.

GTFS Viewer | *React, TypeScript, Vite, JSZip, Papa Parse* Live | GitHub

- Built a browser-based GTFS ZIP viewer with virtual scrolling, pinned headers and colour-coded columns for reviewing large schedule tables across up to three feeds.
- Added feed checks and links from findings to affected rows; kept feed processing local to the browser.

Project Tracker | *TypeScript, React, Firebase Auth, Firestore* Live | GitHub

- Built a project-management application with authenticated, isolated user workspaces and Firestore transactions to keep updates consistent; added backup exports and an access-isolation verification script.

OC Operator Tools | *Node.js, GTFS-RT, PostgreSQL/Supabase* Live | GitHub

- Developed and deployed bus-operator workflows for live block/vehicle lookups, paddle lookup, shuttle schedules and booking boards, integrating Protocol Buffer GTFS-RT feeds with static schedules.
- Implemented normalization and matching for unassigned blocks, stale data and identifier mismatches; added account/saved-work schemas, row-level security, automated tests and a GitHub Actions static-index rebuild workflow.

ERO Operator Tools | *Node.js, Express, Python, PostgreSQL/Supabase* Live | GitHub

- Built paddle lookup, duty status, countdowns and split-duty views around rail operator workflows, with block-to-paddle links in a scheduled Line 1 reference mimic.
- Implemented saved work and access-controlled light rail vehicle defect reporting with relational schemas, migrations, triggers and row-level security policies.

Prime Physio Plus – Clinic Operations System In active clinic use

Next.js, TypeScript, React, PostgreSQL, Prisma, Supabase, Vitest

- Deployed a clinic operations system across **11 separate screens** with a central administration dashboard for patient visits, treatment plans, staff assignments, room status and cleaning; synchronized display updates in real time.
- Implemented relational models, role-based administration and timestamped audit trails; wrote workflow/access-control tests and an illustrated staff manual covering daily administration, reports and operating procedures.

Milano's Pizzeria – Restaurant Operations System GitHub

TypeScript, Vite, Firebase Authentication, Realtime Database, Vitest

- Built a restaurant operations system for drivers, cooks and cashiers, with employee cash-outs, wage calculations, cash reconciliation, receipts and historical pay rates.
- Added supplier and operating-expense tracking, daily sales reconciliation, inventory checklists and shopping lists, and weekly staff scheduling with published team schedules.
- Implemented administrator/device access controls and correction history; wrote model and Firebase security-rule tests for calculations, employee roles and restricted management actions.

Mr. Fez – Online Menu Management and TV Displays Live Site

JavaScript, Firebase Auth/Firestore/Storage, PowerShell, C#/.NET interop

- Built remote menu/price/photo editing, shared drafts, owner-approved publication and automatic delivery to four TV display pages; implemented owner/manager roles, read-only display pairing, revision checks and audit records.
- Developed PowerShell startup/display-mapping scripts with C#/.NET Win32 calls to position dedicated Edge windows; added security-rule and launcher tests in CI and documented installation, recovery and physical-display acceptance checks.